

Association of Mathematics Teachers of India (AMTI)

NMTC-53 -- **SUB-JUNIOR** – (2021 – 2022)

Answer **ALL** Questions. Time: 3 hrs.

- 1) a) In the grid given here, the numbers against each row or column is the sum of all the numbers in that respective row or column.

For example, in the top row,

$$a + b + c + d = 17, \text{ etc.}$$

Find the values of a , b , c and d .

a	b	c	d	17
a	c	c	b	19
c	b	a	b	15
d	d	a	b	15
16	18	18	14	

- b) If $\frac{a}{b} + \frac{b}{a} = x$, $\frac{b}{c} + \frac{c}{b} = y$, $\frac{c}{a} + \frac{a}{c} = z$ find the numerical value of $(x^2 + y^2 + z^2 - xyz)$.

- 2) ABC is a triangle. ℓ_1, ℓ_2, ℓ_3 are three lines that are perpendicular to the sides BC , CA and AB at A_1 , B_1 and C_1 respectively. Prove that ℓ_1, ℓ_2, ℓ_3 are concurrent if and only if $A_1B^2 + C_1A^2 + B_1C^2 = A_1C^2 + C_1B^2 + B_1A^2$

- 3) a) If a, b are real numbers each greater than 1, prove:
 $(1+a)(1+b) < 2(ab + 1)$.

Use this result to prove

$$8(abcd + 1) > (a + 1)(b + 1)(c + 1)(d + 1),$$

where a, b, c, d are real numbers each greater than 1.

- b) The seven digit number $1234 * * *$ is a perfect square number, where each $*$ represents a digit (may be same or distinct). Find the square root of this seven digit number.

Turn over to Page 2

- 4) a) If a, b, c are integers (a, b, c are not zero) such that $a + b + c = 0$, prove that $(2a^4 + 2b^4 + 2c^4)$ is the square of an integer.
- b) Find the smallest positive integer ' n ' such that $4^{27} + 4^{1000} + 4^n$ is a perfect square.
- 5) $PQRS$ is a convex quadrilateral.
Point O is inside the quadrilateral such that $OP = OQ$ and $OS = OR$.
If $\angle POQ = \angle SOR = 120^\circ$ and A, B, C are respectively the mid points of PQ, QR and RS , prove that ABC is an equilateral triangle.